EXTREMAL FUNCTIONS FOR LINEAR FUNCTIONALS ON H^p, $0 < p < 1$, ARE OUTER

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ABSTRACT. Let $C(k,p)$ denote the norm of the functional $f \mapsto f^{(k)}(0)/k!$ on the Hardy space H^p of the unit disc. Brevig and Saksman conjectured that for $0 < p < 1$ the extremal functions for $C(k,p)$ do not vanish in the unit disc, and that $C(k,p)$ is strictly increasing in k. We prove both conjectures by showing that for $0 < p < 1$ every extremal function for a nonzero bounded linear functional on H^p is outer.

1. INTRODUCTION

Let H^p denote the classical Hardy space of analytic functions in the unit disc $\mathbb{D} = \{z \in \mathbb{C} : |z| < 1\}$, which for $0 < p < \infty$ is equipped with

$$\|f\|_{H^p}^p = \lim_{r \to 1^-} \int_0^{2\pi} |f(re^{i\theta})|^p \, \frac{\mathrm{d}\theta}{2\pi}.$$

For an integer $k \geq 1$, let $C(k,p)$ denote the smallest real number such that $|a_k| \leq C(k,p)\|f\|_{H^p}$ holds for every $f(z) = \sum_{n \geq 0} a_n z^n$ in H^p. In other words, $C(k,p)$ is the norm of the bounded linear functional $f \mapsto a_k$ on H^p, and

$$C(k,p) = \sup\Big\{ \operatorname{Re} \frac{f^{(k)}(0)}{k!} : \|f\|_{H^p} = 1 \Big\}. \tag{1}$$

By a standard normal families argument, there are functions attaining the supremum in (1), and we call them extremals for $C(k,p)$.

In the range $1 \leq p < \infty$, it follows from the triangle inequality and Hölder's inequality that $C(k,p) = 1$, and $f(z) = z^k$ is an extremal. For $p = 1$, F. Riesz [7] showed that the extremals of such coefficient problems are polynomials whose zeros in $\mathbb{D}$ are paired with their reflections in the unit circle, so they may vanish in $\mathbb{D}$. In the range $0 < p < 1$, Bondarenko, Brevig, Saksman and Seip [1] computed $C(1,p)$, and Brevig and Saksman [3] computed $C(2,p)$ and $C(3,2/3)$. In each of these cases the extremals do not vanish in $\mathbb{D}$, which led Brevig and Saksman to the following two conjectures [3, Section 5].

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Conjecture 1. *For $0 < p < 1$ any extremal f for $C(k,p)$ does not vanish in $\mathbb{D}$.*

Conjecture 2. *For $0 < p < 1$ the sequence $C(k,p)$ is strictly increasing.*

Note that Conjecture 2 is a consequence of Conjecture 1. Indeed, multiplication by z shows that $C(k,p) \leq C(k+1,p)$. If $C(k,p) = C(k+1,p)$ and f is an extremal for $C(k,p)$, then zf is an extremal for $C(k+1,p)$ which vanishes at the origin (see [3, Section 5]).

Recall that an *extremal* for a bounded linear functional L on H^p is a function f with $\|f\|_{H^p} = 1$ and $\operatorname{Re} L(f) = \|L\|$. Conjecture 1 is the special case $L(f) = f^{(k)}(0)/k!$ of the following result.

Theorem 1. *Suppose that $0 < p < 1$ and that L is a nonzero bounded linear functional on H^p. Then every extremal for L is an outer function.*

To the best of our knowledge, Theorem 1 is new, also for the coefficient functionals. In particular, the extremals for $C(k,p)$ do not vanish in $\mathbb{D}$, and we obtain the following.

Corollary 1. *For $0 < p < 1$ the sequence $C(k,p)$ is strictly increasing.*

Suffridge [8] investigated the extremal problem (1) restricted to functions which do not vanish in $\mathbb{D}$. By Theorem 1, the two problems have the same value and the same extremals when $0 < p < 1$. This confirms a conjecture of Suffridge [8, p. 187]; see also [3, Section 5.2].

Let us briefly describe our approach. As in [1, 3], we factor an extremal as $f = gh^{2/p-1}$, where g and h are in the unit sphere of H^2, the function h is outer, and $g = Ih$ for the inner factor I of f. Brevig and Saksman [3, Lemma 3] used a variational argument, which goes back to F. Riesz [7], to obtain a first-order condition relating g and h. This first-order condition cannot exclude zeros on its own, since it also holds for $p = 1$, where extremals may vanish. We therefore carry the variational argument to second order. In contrast to Riesz and to Brevig and Saksman, we perturb the two factors at once, each in the direction of the other,

$$g \mapsto g + q\varepsilon h, \qquad h \mapsto h + \varepsilon g, \qquad q = \frac{2}{p} - 1,$$

where ε is real. Since the two factors differ precisely by the inner factor I whose presence we wish to exclude, this perturbation is trivial when I is constant. We will see that the relative gain is $q(q-1)\varepsilon^2$ times the variance of $\operatorname{Re} I$ with respect to $|h|^2 \, \mathrm{d}m$ on $\mathbb{T}$, up to $O(\varepsilon^3)$. The gain is a Jensen gap for the function $y \mapsto y^q$, which is convex precisely because $q > 1$, that is, because $p < 1$.

Organization. The present paper is organized into four additional sections. In Section 2 we recall the factorization $f = gh^{2/p-1}$ and the first-order condition of Brevig and Saksman. Section 3 contains the competitor and the proof of Theorem 1. In Section 4 we explain how the argument breaks down when $p \geq 1$, and Section 5 contains some concluding remarks.

2. Preliminaries

Throughout the paper we set $q = 2/p-1$, and we write m for the normalized Lebesgue arc length measure on the unit circle $\mathbb{T} = \{z \in \mathbb{C} : |z| = 1\}$ and $\langle u, v\rangle = \int_{\mathbb{T}} u\overline{v}\,\mathrm{d}m$. Recall that every f in H^p has radial boundary values almost everywhere on $\mathbb{T}$, and that $\|f\|_{H^p}$ is the $L^p(\mathbb{T})$ norm of these boundary values. Suppose that $0 < p < 2$, so that $q > 0$. For $u \in H^2$ and $F \in H^{2/q}$, Hölder's inequality gives

$$\|uF\|_{H^p} \le \|u\|_{H^2}\|F\|_{H^{2/q}}, \tag{2}$$

since $1/p = 1/2 + q/2$. Note also that $\|h^q\|_{H^{2/q}} = \|h\|_{H^2}^q$ for every $h \in H^2$ without zeros in $\mathbb{D}$.

Let L be a nonzero bounded linear functional on H^p with $C = \|L\|$, and let f be an extremal for L. After multiplying f by a unimodular constant, we may assume that $L(f) = C$. Every nonzero function in H^p can be written as $f = IO$, where I is an inner function and O is an outer function. This allows us to factor $f = gh^q$, where

$$h = O^{p/2}, \qquad g = Ih, \qquad w = |h|^2.$$

Here the powers of O are taken with respect to one branch of $\log O$, so that $h^{q+1} = O$. We note that $|g| = |h| = |f|^{p/2}$ on $\mathbb{T}$, which yields $\|g\|_{H^2} = \|h\|_{H^2} = 1$. In particular, $w\,\mathrm{d}m$ is a probability measure on $\mathbb{T}$, and $w > 0$ almost everywhere on $\mathbb{T}$ since h is outer. We write $E(u) = \int_{\mathbb{T}} u\,w\,\mathrm{d}m$ for the average with respect to $w\,\mathrm{d}m$, and $\langle u, v\rangle_w = E(u\overline{v})$ and $\|u\|_w^2 = \langle u, u\rangle_w$ for the inner product and norm of $L^2(w\,\mathrm{d}m)$.

We will repeatedly use two elementary facts. First, since $|I| = 1$ on $\mathbb{T}$, we have $\langle uI, vI\rangle = \langle u, v\rangle$ for $u, v \in L^2(\mathbb{T})$. Second, if ε is real with $|\varepsilon| < 1$, then the principal branch of $(1 + \varepsilon I)^q$ is given by the binomial series

$$(1 + \varepsilon I)^q = \sum_{j\ge 0} \binom{q}{j} \varepsilon^j I^j,$$

which converges uniformly in $\mathbb{D}$, since $|I| \le 1$ there. In particular, $(1+\varepsilon I)^q = 1 + q\varepsilon I + O(\varepsilon^2)$ in H^∞.

The following lemma is the first-order condition of Brevig and Saksman. For $L(f) = f^{(k)}(0)/k!$ it is the equation [3, (9)].

Lemma 1. *Suppose that $0 < p < 2$, and let $f = gh^q$ be as above. Then $L(uh^q) = C\langle u, g\rangle$ for every $u \in H^2$.*

Proof. By (2), the map $u \mapsto L(uh^q)$ is a bounded linear functional on H^2 of norm at most C. Hence $L(uh^q) = \langle u, v\rangle$ for some $v \in H^2$ with $\|v\|_{H^2} \le C$. Since $\langle g, v\rangle = L(f) = C$ and $\|g\|_{H^2} = 1$, we have equality in the Cauchy–Schwarz inequality, and so $v = Cg$. □

In other words, when h is fixed, the best choice of the first factor is g. Indeed, if $\|u\|_{H^2} \le 1$, then Lemma 1 gives $\operatorname{Re} L(uh^q) = C \operatorname{Re}\langle u, g\rangle \le C$, with equality only for $u = g$.

3. The competitor and the proof of Theorem 1

Since $g = Ih$, the two factors in Lemma 1 differ precisely by the inner factor I. This suggests mixing them, and we begin by replacing h with $h + \varepsilon g = (1 + \varepsilon I)h$. This preserves zero-freeness for $|\varepsilon| < 1$, and it reduces to a rescaling of h when I is constant. Our next goal is to determine, to first order, how the optimal first factor responds. Since $(1+\varepsilon I)^q = 1+q\varepsilon I+O(\varepsilon^2)$ in H^∞, Lemma 1 and (2) yield

$$\begin{aligned} L\big(u(1+\varepsilon I)^q h^q\big) &= C\langle u, g\rangle + q\varepsilon C\langle uI, Ih\rangle + O(\varepsilon^2)\|u\|_{H^2} \\ &= C\langle u, g + q\varepsilon h\rangle + O(\varepsilon^2)\|u\|_{H^2} \end{aligned}$$

for real ε and $u \in H^2$, where we used that $\langle uI, Ih\rangle = \langle u, h\rangle$ since $|I| = 1$ on $\mathbb{T}$. Hence, to first order in ε and up to normalization, the optimal direction for the first factor is $g + q\varepsilon h$ (see also Section 5.2). This leads us to the competitor

$$f_\varepsilon = g_\varepsilon h_\varepsilon^q, \qquad g_\varepsilon = g + q\varepsilon h = (I + q\varepsilon)h, \qquad h_\varepsilon = h + \varepsilon g = (1 + \varepsilon I)h,$$

where ε is real and $|\varepsilon| < \min\{1, 1/q\}$. Since $|\varepsilon I| < 1$ in $\mathbb{D}$, the function h_ε has no zeros in $\mathbb{D}$, and we put $h_\varepsilon^q = (1 + \varepsilon I)^q h^q$ with the principal branch of $(1 + \varepsilon I)^q$. Note that the competitor reduces to the factorization $f = gh^q$ when $\varepsilon = 0$. On $\mathbb{T}$ we set

$$A = (1 + \varepsilon I)^q, \qquad B = 1 + q\varepsilon I, \qquad R(\varepsilon) = \frac{\operatorname{Re}\langle A, B\rangle_w}{\|B\|_w \|1 + \varepsilon I\|_w^q},$$

where the denominator is positive since $|q\varepsilon| < 1$.

Lemma 2. *Suppose that $0 < p < 2$ and that ε is real with $|\varepsilon| < \min\{1, 1/q\}$. Then*

$$\operatorname{Re} L(f_\varepsilon) = CR(\varepsilon)\|g_\varepsilon\|_{H^2}\|h_\varepsilon\|_{H^2}^q,$$

and in particular $R(\varepsilon) \le 1$.

Proof. We begin by noting that $(I + q\varepsilon)\overline{I} = \overline{B}$ on $\mathbb{T}$, since $|I| = 1$ on $\mathbb{T}$ and ε is real. Taking moduli and using that $|h|^2 = w$, we find that $\|g_\varepsilon\|_{H^2} = \|B\|_w$ and $\|h_\varepsilon\|_{H^2} = \|1+\varepsilon I\|_w$. Next, we write $f_\varepsilon = uh^q$ with $u = (I+q\varepsilon)Ah \in H^2$. Since $g = Ih$, Lemma 1 gives

$$L(f_\varepsilon) = C\langle (I + q\varepsilon)Ah, Ih\rangle = C\int_{\mathbb{T}} A\,(I + q\varepsilon)\overline{I}\, w \,\mathrm{d}m = C\langle A, B\rangle_w,$$

and taking real parts we obtain the stated identity. Since $\|L\| = C$, it follows from (2) that

$$\operatorname{Re} L(f_\varepsilon) \le C\|f_\varepsilon\|_{H^p} \le C\|g_\varepsilon\|_{H^2}\|h_\varepsilon\|_{H^2}^q,$$

and comparing this with the identity, we conclude that $R(\varepsilon) \le 1$. □

Lemma 2 reduces Theorem 1 to an expansion of $R(\varepsilon)$, which is the content of the next lemma. Here and in what follows, $V = E((\operatorname{Re} I)^2) - E(\operatorname{Re} I)^2$ denotes the variance of $\operatorname{Re} I$ with respect to the probability measure $w\,\mathrm{d}m$.

Lemma 3. *Suppose that $0 < p < 2$. Then*

$$R(\varepsilon) = 1 + q(q-1)V\varepsilon^2 + O(\varepsilon^3) \tag{3}$$

as $\varepsilon \to 0$.

Proof. Set $Y = |1 + \varepsilon I|^2$. Since $|A|^2 = Y^q$ and $\|1 + \varepsilon I\|_w^2 = E(Y)$, the ratio splits as

$$R(\varepsilon) = \frac{\operatorname{Re}\langle A, B\rangle_w}{\|A\|_w \|B\|_w} \cdot \frac{E(Y^q)^{1/2}}{E(Y)^{q/2}}. \tag{4}$$

The first factor is the cosine of the angle between A and B in $L^2(w\,\mathrm{d}m)$. For unit vectors a and b we have $\operatorname{Re}\langle a, b\rangle_w = 1 - \|a-b\|_w^2/2$. Since $|\varepsilon I| = |\varepsilon|$ on $\mathbb{T}$, the binomial series yields $A = B + O(\varepsilon^2)$ uniformly on $\mathbb{T}$. As $\|B\|_w \geq 1 - q|\varepsilon|$, we get

$$\Big\| \frac{A}{\|A\|_w} - \frac{B}{\|B\|_w} \Big\|_w \leq \frac{2\|A - B\|_w}{\|B\|_w} = O(\varepsilon^2),$$

and so the first factor is $1 - O(\varepsilon^4)$.

The second factor is a Jensen ratio for the function $y \mapsto y^q$. Taylor's formula gives $y^q = 1 + q(y-1) + q(q-1)(y-1)^2/2 + O(|y-1|^3)$ uniformly for $|y - 1| \leq 1/2$. Since $Y = 1 + 2\varepsilon \operatorname{Re} I + \varepsilon^2$, we have $|Y - 1| \leq 3|\varepsilon|$, and so for small ε

$$E(Y^q) = 1 + q\big(E(Y) - 1\big) + \binom{q}{2} E\big((Y-1)^2\big) + O(\varepsilon^3),$$

$$E(Y)^q = 1 + q\big(E(Y) - 1\big) + \binom{q}{2} \big(E(Y) - 1\big)^2 + O(\varepsilon^3).$$

The difference of the two quadratic terms is $q(q-1)/2$ times the variance of Y, which equals $4\varepsilon^2 V$ since $Y - 1 - \varepsilon^2 = 2\varepsilon \operatorname{Re} I$. Hence

$$E(Y^q) - E(Y)^q = 2q(q-1)V\varepsilon^2 + O(\varepsilon^3).$$

As $E(Y) = 1 + O(\varepsilon)$, the second factor is

$$\Big(1 + 2q(q-1)V\varepsilon^2 + O(\varepsilon^3)\Big)^{1/2} = 1 + q(q-1)V\varepsilon^2 + O(\varepsilon^3),$$

and multiplying the two factors we obtain (3). □

Proof of Theorem 1. Suppose that I is not constant. By Lemmas 2 and 3, we have $q(q-1)V\varepsilon^2 + O(\varepsilon^3) \leq 0$ for every small real ε, and hence $q(q-1)V \leq 0$. Since $p < 1$, we have $q > 1$, and so $V = 0$. This means that $\operatorname{Re} I$ is constant almost everywhere with respect to $w\,\mathrm{d}m$, and hence almost everywhere on $\mathbb{T}$, since $w > 0$ almost everywhere. A bounded analytic function whose real part has constant boundary values is constant, so I is constant. This is a contradiction. □

4. The case $p \geq 1$

Theorem 1 does not hold for any $1 \leq p < \infty$. Indeed, in this range $C(k,p) = 1$, and $f(z) = z^k$ is an extremal for $C(k,p)$ which vanishes at the origin. It is instructive to see how this is reflected in the argument. Apart from the restriction $p < 2$, which (2) requires, the proof uses $p < 1$ in one place only, namely the sign of $q(q-1)$ in (3).

When $1 \leq p < 2$, we have $0 < q \leq 1$, and (2) and Lemmas 1 to 3 hold with the same proofs. However, since $q(q-1) \leq 0$, the inequality $R(\varepsilon) \leq 1$ now agrees with (3) and gives no contradiction.

The endpoint $p = 1$ shows what the competitor does instead. Here $q = 1$, so $g_\varepsilon = (I+\varepsilon)h$ and $h_\varepsilon = (1+\varepsilon I)h$ have the same modulus on $\mathbb{T}$. Consequently, the numerator and the denominator of $R(\varepsilon)$ are both equal to $\|1+\varepsilon I\|_w^2$, and $R(\varepsilon) = 1$ for every $|\varepsilon| < 1$. By Lemma 2 and (2), we find that

$$\operatorname{Re} L(f_\varepsilon) = C\|g_\varepsilon\|_{H^2}\|h_\varepsilon\|_{H^2} \geq C\|f_\varepsilon\|_{H^1},$$

so $f_\varepsilon/\|f_\varepsilon\|_{H^1}$ is again an extremal. In other words, the perturbation moves along a curve of extremals. For $L(f) = f'(0)$ and $f(z) = z$, this is the curve $(z+\varepsilon)(1+\varepsilon z)/(1+\varepsilon^2)$, whose zero $-\varepsilon$ in $\mathbb{D}$ is paired with the zero $-1/\varepsilon$ outside $\overline{\mathbb{D}}$, in accordance with Riesz's description [7].

When $p \geq 2$, we have $q \leq 0$ and the argument has no counterpart. For $p = 2$ we have $f = g$, so there is nothing to perturb, and for $p > 2$ the exponent $2/q$ in (2) is negative.

5. Concluding remarks

5.1. **The choice of the first factor.** For any first factor $Xh \in H^2$, the proof of Lemma 2 gives

$$\frac{\operatorname{Re} L(Xh\, h_\varepsilon^q)}{C\|Xh\|_{H^2}\|h_\varepsilon\|_{H^2}^q} = \frac{\operatorname{Re}\langle A, \overline{X}I\rangle_w}{\|A\|_w\|\overline{X}I\|_w} \cdot \frac{E(Y^q)^{1/2}}{E(Y)^{q/2}}.$$

Only the cosine depends on the first factor. Choosing the first factor optimal for h_ε, by the Cauchy–Schwarz inequality as in [3], therefore maximizes the cosine. Our choice $X = I + q\varepsilon$ gives $\overline{X}I = B$, for which the cosine is already $1 - O(\varepsilon^4)$, so the optimal first factor does not change the second-order term. Computing it would require the Riesz representative of $u \mapsto L(uh_\varepsilon^q)$ on H^2, and hence the projection P_+ from Section 5.2, while the explicit competitor only uses the representative Cg from Lemma 1.

To see how sensitive the cosine is to this choice, take $X = I + \delta$ with δ real, so that $Xh = g + \delta h$ and $\overline{X}I = 1 + \delta I$. Set $m_1 = E(\operatorname{Re} I)$. Since $(1+tI)/\|1+tI\|_w = 1 + t(I - m_1) + O(t^2)$ and $\|I - m_1\|_w^2 = 1 - m_1^2$, we find that the ratio above equals

$$1 + q(q-1)V\varepsilon^2 - \frac{1-m_1^2}{2}(\delta - q\varepsilon)^2 + O\big((|\varepsilon| + |\delta|)^3\big).$$

If I is not constant, then $|m_1| < 1$, since $w > 0$ almost everywhere, so $\delta = q\varepsilon$ is optimal to second order. For $q = 1$ and $\delta = \varepsilon$ we have $A = B$ and

$Y^q = Y$, so both factors in (4) are equal to 1. This is the identity $R(\varepsilon) = 1$ from Section 4.

5.2. **The direction of the perturbation.** The direction $\varphi = I$ in Section 3 can also be understood as follows. For a perturbation $h \mapsto (1 + \varepsilon\varphi)h$ with $\varphi \in H^\infty$, the computation in Section 3 shows that the linear response of the first factor is $g + q\varepsilon P_+(\overline{\varphi} g)$, where P_+ is the orthogonal projection of $L^2(\mathbb{T})$ onto H^2. The part of $\overline{\varphi} g$ removed by P_+ does not contribute to this first-order response, and for $\varphi = I$ nothing is removed, since $\overline{I} g = h$.

5.3. **A prescribed value at the origin.** Brevig, Grepstad and Instanes [2] considered the extremal problem

$$\Phi_k(p, t) = \sup\Big\{ \operatorname{Re} \frac{f^{(k)}(0)}{k!} : \|f\|_{H^p} \le 1 \text{ and } f(0) = t \Big\}$$

for $0 \le t \le 1$. For $k = 1$ and $0 < p < 1$ this problem was studied by Kabaila [5, 6] and Connelly [4], and solved completely in [2, Theorem 12]. There is $0 < t_p < 1/2$ such that the extremal has one zero in $\mathbb{D}$ when $t < t_p$ and none when $t > t_p$, and Kabaila showed that t_p is the only value of t for which there are two extremals, one with a zero and one without. Hence, with a second condition on the coefficients, extremals may vanish in $\mathbb{D}$ also for $0 < p < 1$. The proof of Theorem 1 does not apply to this problem, since the competitor f_ε does not preserve the condition $f(0) = t$.

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